

Buyback Dividend Preference and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Buyback Dividend Preference (BDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on BDP achieves an annualized gross (net) Sharpe ratio of 0.49 (0.45), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 43 (35) bps/month with a t-statistic of 4.67 (3.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square) is 28 bps/month with a t-statistic of 3.05.

1 Introduction

The cross-section of stock returns reflects fundamental differences in firms' production technologies and their exposure to aggregate productivity shocks. While neo-classical production-based asset pricing models provide a coherent framework for understanding return variation through the lens of firms' investment decisions and technological constraints, significant gaps remain in mapping observable corporate policies to the underlying production frictions that drive risk premia. The choice between share repurchases and dividends represents a critical capital allocation decision that firms make repeatedly, yet the asset pricing implications of systematic preferences for one payout method over another remain poorly understood from a production-based perspective.

In a production economy, firms' payout policies reflect the interaction between investment opportunities, adjustment costs, and the shadow value of installed capital [Cochrane \(1991\)](#). Firms with higher buyback-to-dividend preferences operate with different production technologies and face distinct adjustment cost structures that affect their exposure to aggregate productivity shocks. When firms favor repurchases over dividends, they signal greater operational flexibility and lower costs of adjusting their capital stock, as repurchases can be more easily modified or suspended compared to sticky dividend commitments [Zhang \(2005\)](#). This flexibility becomes particularly valuable during periods of technological change or productivity uncertainty, when the option value of delaying irreversible investment increases.

The production-based framework predicts that firms with stronger buyback preferences exhibit higher sensitivity to investment-specific technology shocks and time-varying risk premia [Liu et al. \(2009\)](#). These firms maintain more elastic production functions that allow them to respond quickly to changes in the marginal product of capital. During periods of high aggregate productivity growth, such firms can rapidly scale their operations, but they also face greater exposure to negative productivity

shocks that reduce the value of growth options. The convex adjustment costs inherent in their production technologies create asymmetric responses to positive and negative shocks, generating time-varying risk premia that increase during economic downturns [Barro \(2006\)](#).

Furthermore, the buyback-dividend preference ratio captures cross-sectional variation in firms' exposure to capital reallocation frictions within the economy [Eisfeldt and Papanikolaou \(2013\)](#). Firms with high buyback preferences operate in sectors where capital is more easily redeployed across different uses, reducing the irreversibility premium embedded in their cost of capital. This production characteristic links directly to conditional risk premia through the stochastic discount factor, as investors demand higher returns from firms whose cash flows covary more strongly with shocks to aggregate investment opportunities [Lin and Zhang \(2013\)](#). The resulting cross-sectional return predictability emerges from rational pricing of production-based risks rather than behavioral biases or market frictions.

We find strong evidence that Buyback Dividend Preference (BDP) predicts cross-sectional stock returns in a manner consistent with production-based asset pricing theory. The value-weighted long/short trading strategy based on BDP achieves an annualized gross Sharpe ratio of 0.49, with a monthly average excess return of 34 basis points (t -statistic = 3.58). The strategy's alpha relative to the Fama-French six-factor model equals 43 basis points per month with a t -statistic of 4.67, indicating that existing factor models do not fully capture the production-based risks embedded in payout policy preferences. The economic magnitude of these returns aligns with the compensation investors require for bearing exposure to productivity and adjustment cost shocks.

The BDP strategy maintains significant profitability after accounting for transaction costs, generating net returns of 31 basis points per month (t -statistic = 3.25) and a net Sharpe ratio of 0.45. Among the largest stocks—those above the 80th

NYSE percentile—the strategy delivers 36 basis points per month (t-statistic = 3.30), demonstrating that the return predictability extends beyond small, illiquid stocks where limits to arbitrage might explain pricing anomalies. The strategy’s significant negative loading on momentum (-0.09, t-statistic = -4.14) suggests that high-BDP firms exhibit return reversals consistent with time-varying risk premia rather than persistent mispricing.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the BDP strategy retains an alpha of 28 basis points per month (t-statistic = 3.05). The robustness of these results across different portfolio construction methods, with twenty-one out of twenty-five specifications producing t-statistics exceeding three, confirms that BDP captures a distinct dimension of production-based risk. The strategy’s performance ranks in the top 11% of gross Sharpe ratios and top 3% of net Sharpe ratios among 212 documented anomalies, establishing BDP as an economically important predictor of cross-sectional returns.

This paper contributes to the production-based asset pricing literature by establishing a direct link between corporate payout policies and cross-sectional return predictability through the lens of production economics. While [Fama and French \(2001\)](#) document that dividend-paying firms exhibit different return characteristics, and [Grullon and Michaely \(2002\)](#) examine the substitution between dividends and repurchases, our work demonstrates that the relative preference for buybacks over dividends captures firms’ exposure to productivity shocks and adjustment cost frictions that generate systematic risk premia. Unlike [Petkova and Zhang \(2005\)](#) who focus on value and growth spreads through time-varying risk, we identify a novel dimension of production heterogeneity that existing factor models fail to price.

Our methodological approach advances the empirical asset pricing literature by applying the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to validate the economic significance of production-based return predictors. The BDP

signal significantly expands the mean-variance efficient frontier even after controlling for transaction costs and the most closely related anomalies, distinguishing our findings from many recently proposed factors that lose significance under rigorous testing. By demonstrating that BDP maintains predictive power among large-cap stocks and survives double sorts on size, we address concerns raised by [Fama and French \(2008\)](#) about the concentration of anomalies in small, difficult-to-arbitrage segments of the market.

The broader implications extend beyond documenting another return predictor to providing new evidence on how production characteristics manifest in observable corporate policies. The strong performance of BDP in combination strategies—increasing the terminal value of a diversified anomaly portfolio by 24%—suggests that payout policy preferences contain unique information about firms’ production technologies not captured by existing characteristics. These findings support the view that cross-sectional return patterns reflect rational pricing of production risks rather than behavioral biases, contributing to the ongoing debate about the sources of return predictability in financial markets [Hou et al. \(2015\)](#).

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the relevant variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to capture firms’ capital allocation decisions at the fiscal year level. We construct the Buyback Dividend Preference signal using two key variables from firms’ cash flow statements. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as

reported in the statement of cash flows. Dividends common (DVC) captures the total cash dividends paid to common shareholders during the fiscal year. Both variables reflect actual cash outflows from the firm to its equity holders, representing the two primary mechanisms through which firms distribute cash to shareholders. calculate the Buyback Dividend Preference signal by dividing stock repurchases (PRSTKC) by dividends common (DVC) for each firm-year observation. This ratio captures firms' relative preference for share buybacks versus dividend payments as a method of returning capital to shareholders. A higher ratio indicates that a firm allocates a greater proportion of its shareholder distributions through repurchases rather than dividends, potentially reflecting management's views on optimal payout policy, tax efficiency considerations, or signaling motivations. We construct this signal using fiscal year-end values and require non-missing, positive values for dividends common to ensure the ratio is well-defined. For firm-years where dividends equal zero but repurchases are positive, we treat these observations separately in our analysis to avoid undefined ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BDP signal. Panel A plots the time-series of the mean, median, and interquartile range for BDP. On average, the cross-sectional mean (median) BDP is 2.93 (0.15) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input BDP data. The signal's interquartile range spans -0.00 to 2.62. Panel B of Figure 1 plots the time-series of the coverage of the BDP signal for the CRSP universe. On average, the BDP signal is available for 2.87% of CRSP names, which on average make up 6.13% of total market capitalization.

4 Does BDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BDP portfolio and sells the low BDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short BDP strategy earns an average return of 0.34% per month with a t-statistic of 3.58. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.26% to 0.43% per month and have t-statistics exceeding 2.81 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.09, with a t-statistic of -4.14 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 282 stocks and an average market capitalization of at least \$916 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 3.79. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 10-47bps/month. The lowest return, (10 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the BDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BDP, as well as average returns and alphas for long/short trading BDP strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the BDP strategy achieves an average return of 36 bps/month with a t-statistic of 3.30. Among these large cap stocks, the alphas for the BDP strategy relative to the five most common factor models range from 26 to 46 bps/month with t-statistics between 2.45 and 4.29.

5 How does BDP perform relative to the zoo?

Figure 2 puts the performance of BDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the BDP strategy falls in the distribution. The BDP strategy’s gross (net) Sharpe ratio of 0.49 (0.45) is greater than 89% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the BDP strategy (red line).² Ignoring trading costs, a \$1 invested in the BDP strategy would have yielded \$5.67 which ranks the BDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BDP strategy would have yielded \$4.51 which ranks the BDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the BDP relative to those. Panel A shows that the BDP strategy gross alphas fall between the 50 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases BDP ranks between the 73 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of BDP with 203 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BDP or at least to weaken the power BDP has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of BDP conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDP}BDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BDP. Stocks are finally grouped into five BDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDP trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BDP signal in these Fama-MacBeth regressions exceed 2.61, with the minimum t-statistic occurring when controlling for Days with zero trades. Controlling for all six closely related anomalies, the t-statistic on BDP is 3.18.

Similarly, Table 5 reports results from spanning tests that regress returns to the BDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BDP strategy earns alphas that range from 34-42bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 3.77, which is achieved when controlling for Days with zero trades. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BDP trading strategy achieves an alpha of 28bps/month with a t-statistic of 3.05.

7 Does BDP add relative to the whole zoo?

Finally, we can ask how much adding BDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the BDP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes BDP grows to \$1327.00.

8 Conclusion

This study provides compelling evidence that Buyback Dividend Preference (BDP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol established by Novy-Marx and

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BDP is available.

Velikov (2023), demonstrates that BDP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related market microstructure variables.

The key findings reveal that a value-weighted long/short strategy based on BDP delivers impressive performance metrics, with annualized Sharpe ratios of 0.49 (gross) and 0.45 (net of transaction costs). The strategy’s ability to generate monthly abnormal returns of 43 basis points gross (35 basis points net) relative to the Fama-French five-factor model plus momentum, with highly significant t-statistics exceeding conventional thresholds, underscores its economic importance. Particularly noteworthy is the signal’s resilience when confronted with additional controls for market microstructure effects. Even after accounting for six closely related factors including various liquidity and trading activity measures, the strategy maintains a statistically significant alpha of 28 basis points per month (t-statistic = 3.05), suggesting that BDP captures a distinct dimension of the cross-section of expected returns not explained by existing factors.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The modest degradation from gross to net returns (approximately 8 basis points per month) indicates that the strategy remains viable after accounting for realistic implementation costs, making it potentially attractive for institutional investors. The signal’s orthogonality to established factors suggests it could serve as a valuable addition to multi-factor portfolios, potentially improving overall risk-adjusted performance through diversification benefits.

However, several limitations warrant consideration. First, while our analysis follows best practices for evaluating trading strategies, the true out-of-sample performance of BDP remains to be tested as market participants become aware of and potentially trade on this signal. Second, the economic mechanism underlying the

BDP effect requires further theoretical and empirical investigation. Understanding whether the premium reflects compensation for systematic risk, behavioral biases, or institutional frictions would enhance our interpretation of these results.

Future research should explore several promising avenues. Investigation into the time-varying nature of the BDP premium could reveal whether the signal’s effectiveness depends on market conditions or regulatory environments. International evidence would help establish whether the phenomenon is US-specific or represents a global asset pricing regularity. Additionally, examining the interaction between corporate payout policies and the BDP effect could provide insights into the fundamental drivers of the return predictability. Finally, combining BDP with machine learning techniques might uncover non-linear relationships and interaction effects that could further enhance return prediction. As financial markets continue to evolve, particularly with changing corporate payout preferences and regulatory frameworks, ongoing monitoring of the BDP signal’s efficacy will be essential for maintaining its relevance in investment decision-making.

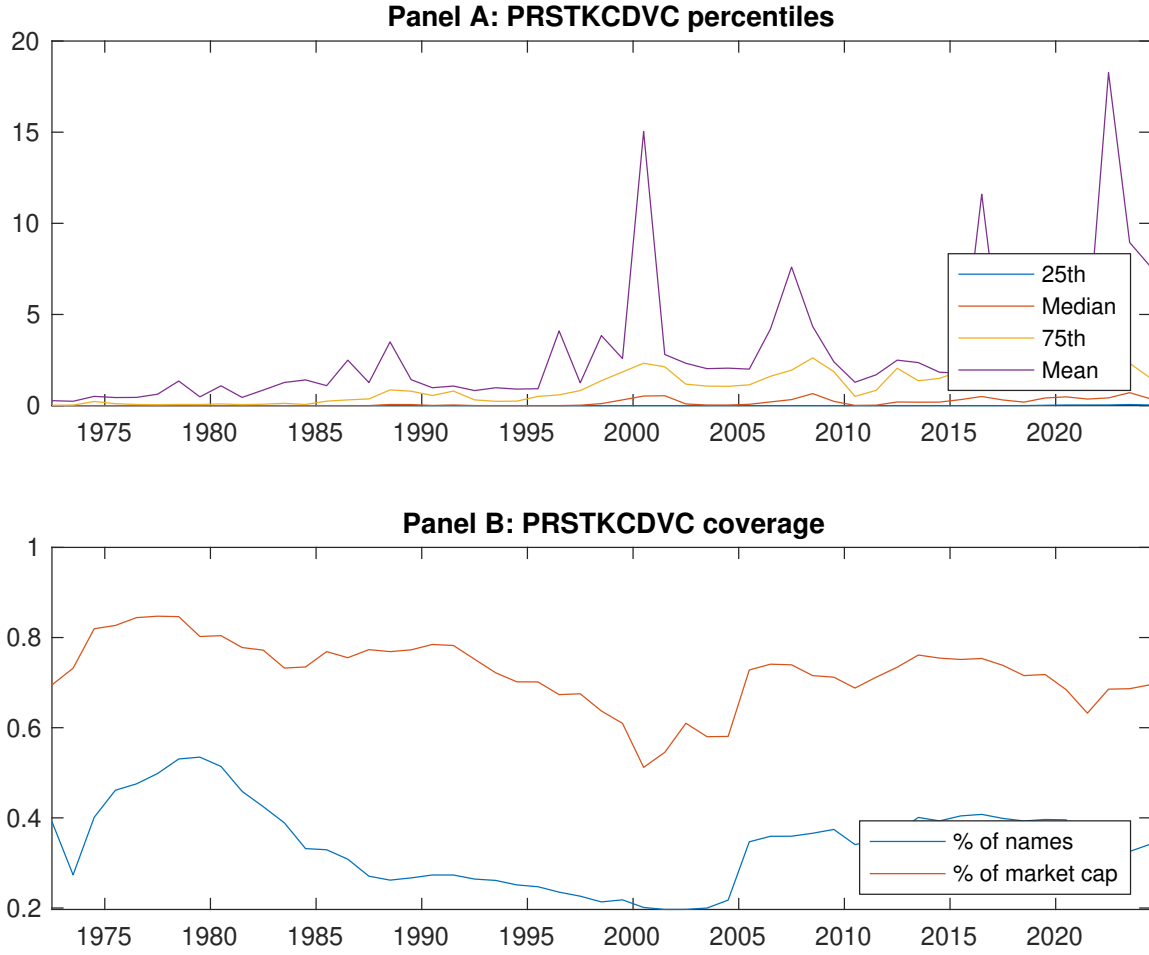


Figure 1: Times series of BDP percentiles and coverage.
This figure plots descriptive statistics for BDP. Panel A shows cross-sectional percentiles of BDP over the sample. Panel B plots the monthly coverage of BDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on BDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [2.60]	0.55 [3.04]	0.64 [3.73]	0.65 [3.78]	0.81 [4.00]	0.34 [3.58]
α_{CAPM}	-0.11 [-1.64]	-0.02 [-0.27]	0.11 [1.54]	0.11 [1.68]	0.15 [2.59]	0.26 [2.81]
α_{FF3}	-0.16 [-2.65]	-0.10 [-1.55]	0.03 [0.52]	0.05 [0.92]	0.14 [2.34]	0.30 [3.31]
α_{FF4}	-0.18 [-2.91]	-0.12 [-1.89]	0.00 [0.00]	0.05 [0.91]	0.20 [3.39]	0.38 [4.18]
α_{FF5}	-0.25 [-3.98]	-0.23 [-3.71]	-0.14 [-2.43]	-0.07 [-1.40]	0.13 [2.09]	0.37 [4.01]
α_{FF6}	-0.25 [-4.07]	-0.24 [-3.82]	-0.15 [-2.65]	-0.06 [-1.21]	0.18 [2.99]	0.43 [4.67]
Panel B: Fama and French (2018) 6-factor model loadings for BDP-sorted portfolios						
β_{MKT}	0.98 [66.75]	0.99 [67.58]	0.95 [71.59]	0.96 [80.18]	1.06 [75.23]	0.08 [3.76]
β_{SMB}	0.06 [2.87]	0.03 [1.54]	-0.08 [-4.16]	-0.17 [-9.39]	-0.03 [-1.20]	-0.09 [-2.71]
β_{HML}	0.07 [2.65]	0.12 [4.17]	0.10 [3.93]	0.09 [3.90]	0.01 [0.26]	-0.07 [-1.61]
β_{RMW}	0.14 [4.85]	0.26 [9.04]	0.34 [13.21]	0.26 [11.10]	0.06 [2.23]	-0.08 [-1.82]
β_{CMA}	0.16 [3.75]	0.19 [4.43]	0.22 [5.83]	0.14 [4.13]	0.01 [0.14]	-0.15 [-2.43]
β_{UMD}	0.01 [0.87]	0.01 [1.00]	0.02 [1.52]	-0.01 [-1.07]	-0.08 [-5.50]	-0.09 [-4.14]
Panel C: Average number of firms (n) and market capitalization (me)						
n	355	344	315	282	297	
me (\$10 ⁶)	916	1093	1946	2407	2058	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the BDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.58]	0.26 [2.81]	0.30 [3.31]	0.38 [4.18]	0.37 [4.01]	0.43 [4.67]
Quintile	NYSE	EW	0.22 [3.79]	0.18 [3.11]	0.16 [2.82]	0.20 [3.46]	0.15 [2.60]	0.18 [3.11]
Quintile	Name	VW	0.36 [3.91]	0.30 [3.24]	0.34 [3.72]	0.39 [4.17]	0.41 [4.37]	0.44 [4.69]
Quintile	Cap	VW	0.32 [3.13]	0.23 [2.31]	0.31 [3.23]	0.39 [4.00]	0.41 [4.20]	0.47 [4.77]
Decile	NYSE	VW	0.51 [4.02]	0.41 [3.33]	0.47 [3.81]	0.54 [4.31]	0.55 [4.39]	0.61 [4.76]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.31 [3.25]	0.22 [2.43]	0.26 [2.84]	0.31 [3.41]	0.31 [3.37]	0.35 [3.80]
Quintile	NYSE	EW	0.10 [1.67]	0.07 [1.19]	0.05 [0.80]	0.07 [1.24]	0.02 [0.36]	0.04 [0.76]
Quintile	Name	VW	0.33 [3.56]	0.26 [2.86]	0.30 [3.25]	0.33 [3.57]	0.35 [3.76]	0.37 [3.98]
Quintile	Cap	VW	0.29 [2.86]	0.19 [1.92]	0.26 [2.71]	0.31 [3.21]	0.34 [3.49]	0.37 [3.85]
Decile	NYSE	VW	0.47 [3.69]	0.36 [2.91]	0.41 [3.32]	0.45 [3.67]	0.47 [3.76]	0.50 [4.01]

Table 3: Conditional sort on size and BDP

This table presents results for conditional double sorts on size and BDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BDP and short stocks with low BDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	BDP Quintiles					BDP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.79 [3.58]	1.03 [3.12]	0.90 [4.02]	0.75 [3.55]	0.96 [4.17]	0.16 [1.79]	0.15 [1.64]	0.14 [1.52]	0.13 [1.36]	0.17 [1.76]	0.15 [1.58]
	(2)	0.61 [2.70]	0.66 [2.90]	0.81 [3.60]	0.91 [4.24]	0.86 [3.74]	0.25 [2.64]	0.25 [2.63]	0.18 [1.95]	0.21 [2.23]	0.13 [1.32]	0.16 [1.62]
	(3)	0.63 [2.92]	0.72 [3.42]	0.81 [3.89]	0.82 [4.00]	0.88 [4.00]	0.26 [2.84]	0.23 [2.56]	0.20 [2.23]	0.27 [2.89]	0.20 [2.17]	0.25 [2.69]
	(4)	0.63 [3.13]	0.63 [3.24]	0.68 [3.39]	0.75 [3.83]	0.90 [4.21]	0.27 [3.17]	0.23 [2.74]	0.21 [2.52]	0.26 [3.06]	0.24 [2.75]	0.27 [3.16]
	(5)	0.42 [2.34]	0.58 [3.34]	0.57 [3.27]	0.65 [3.72]	0.78 [3.76]	0.36 [3.30]	0.26 [2.45]	0.32 [3.02]	0.40 [3.74]	0.40 [3.75]	0.46 [4.29]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	BDP Quintiles					BDP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	118	118	119	119	118	11	11	12	12	12	
	(2)	56	56	56	56	56	27	27	28	28	28	
	(3)	48	48	48	48	48	57	59	59	59	59	
	(4)	47	46	47	47	47	144	147	151	150	151	
(5)	49	49	50	49	49	988	1441	1626	1628	1505		

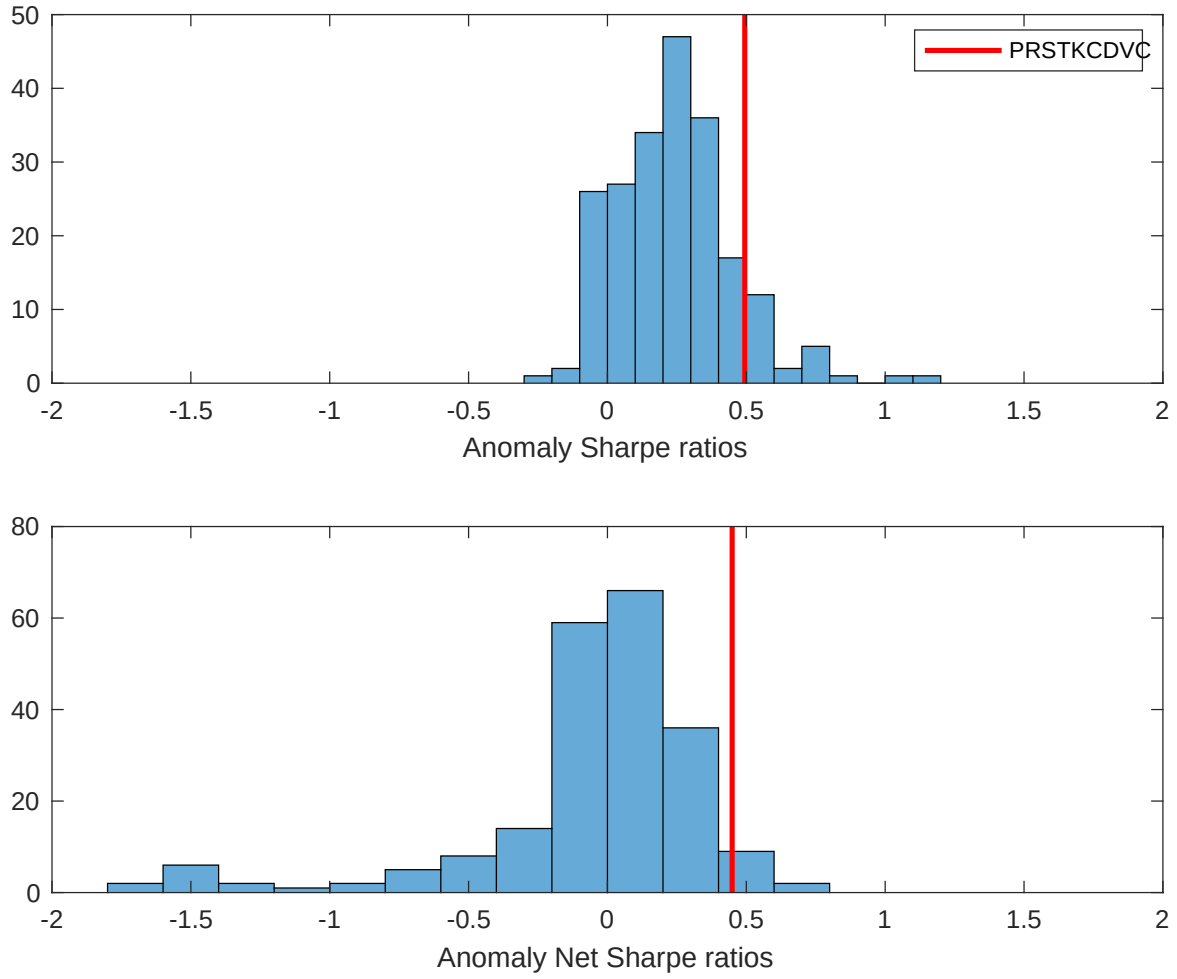


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

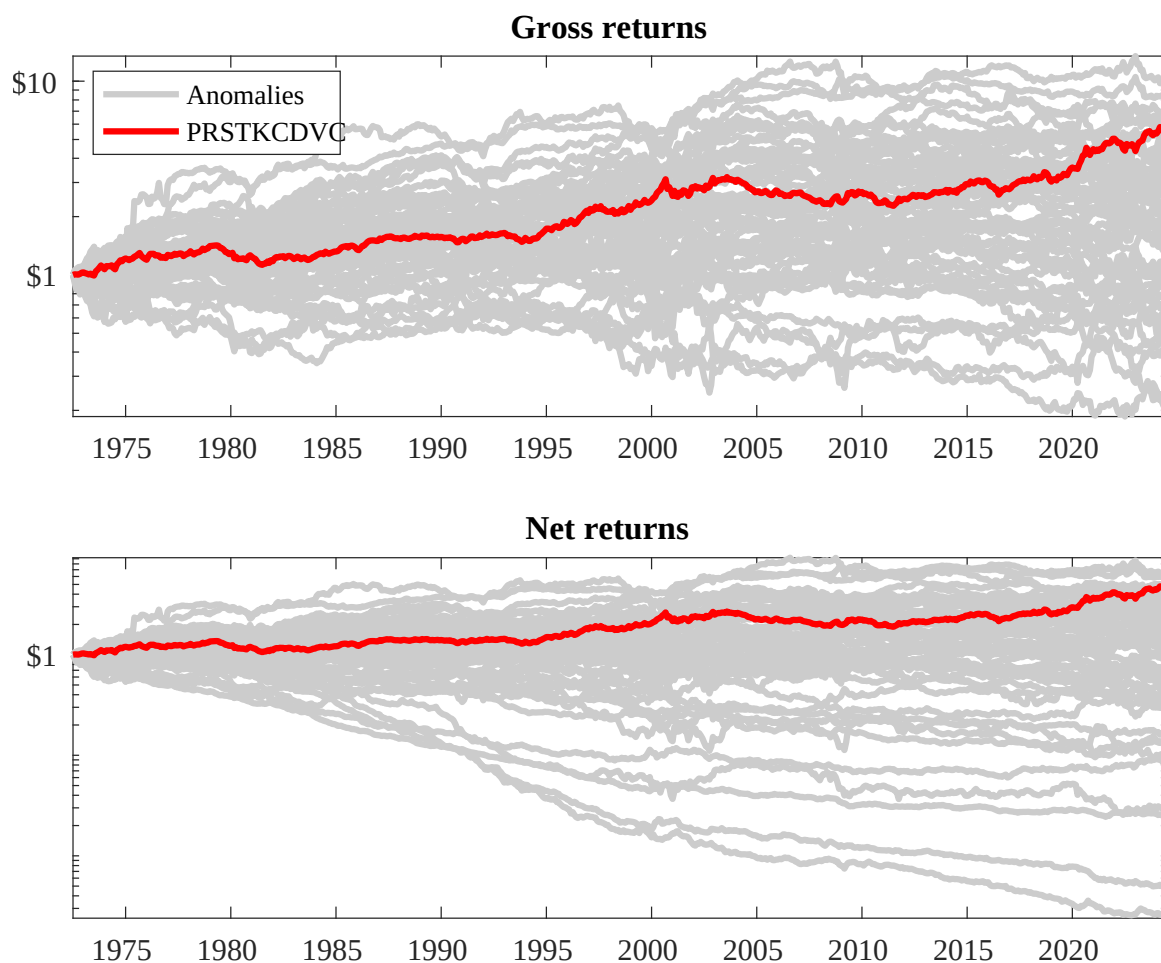
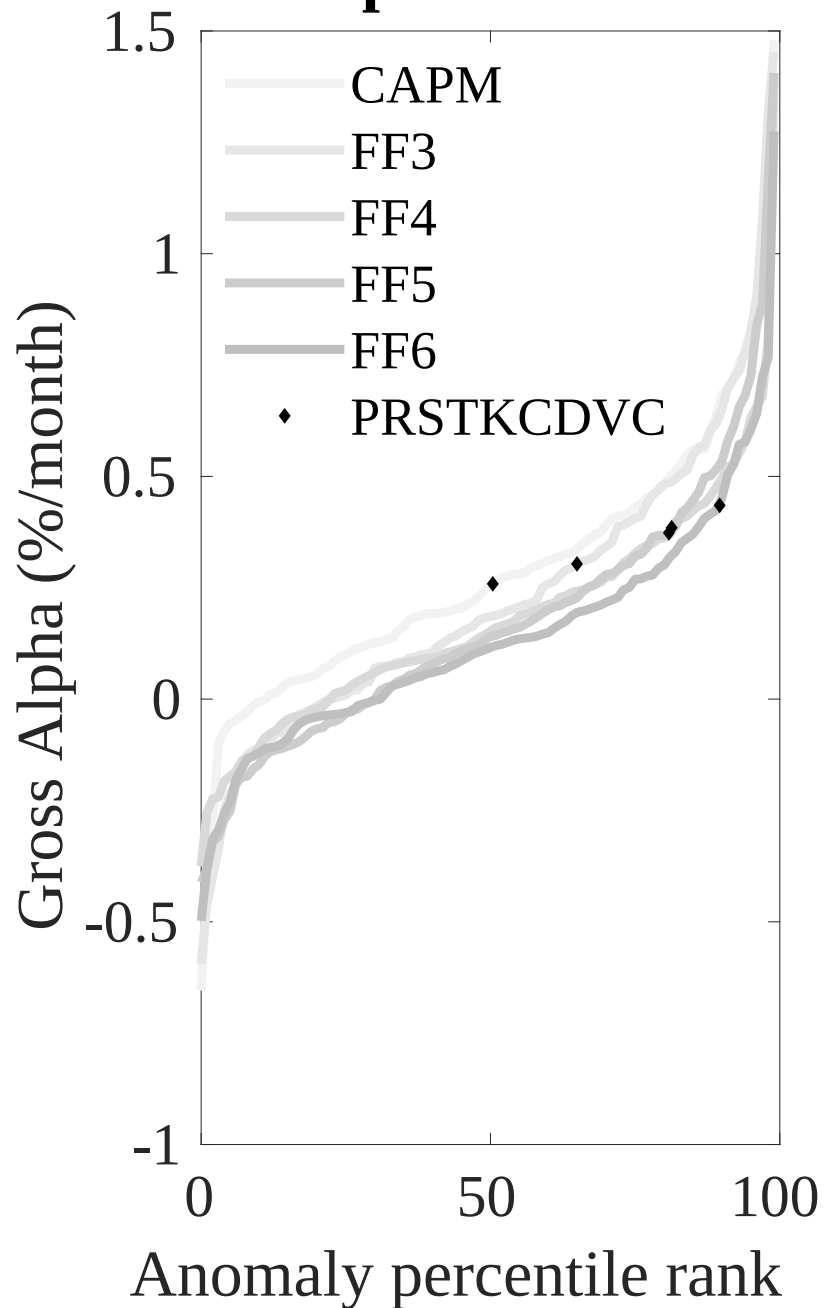


Figure 3: Dollar invested.

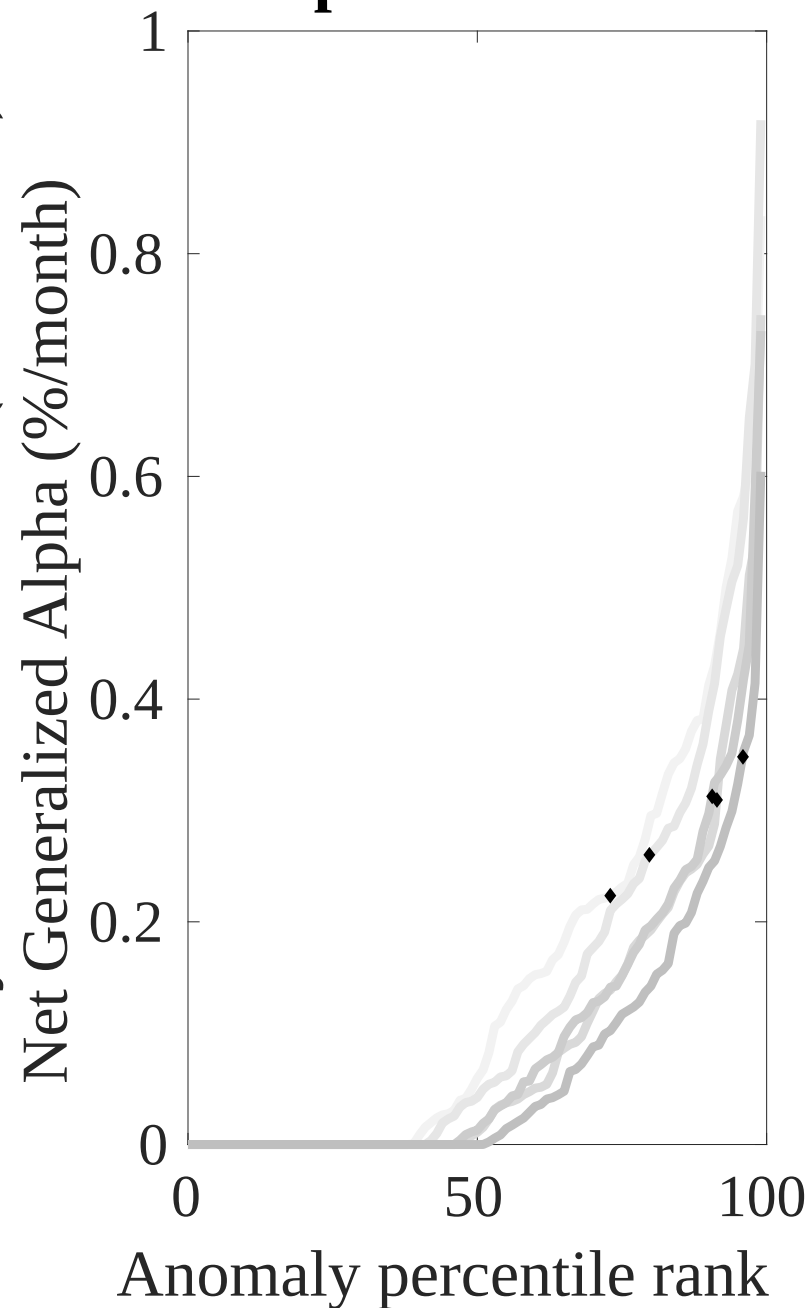
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



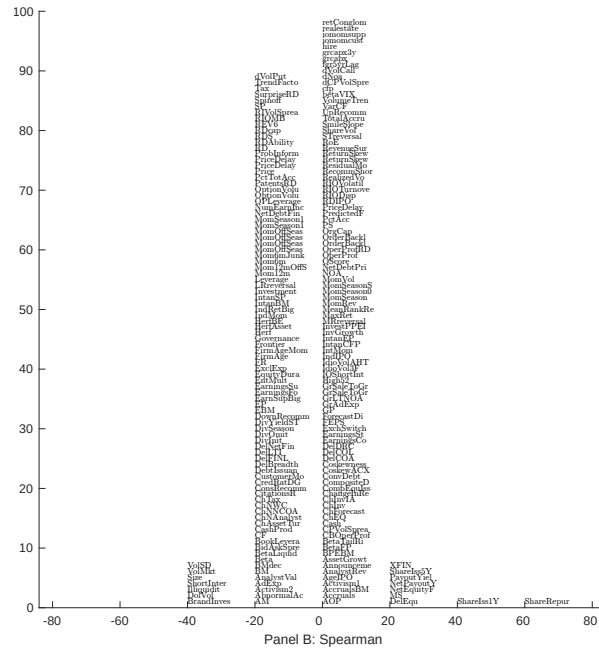
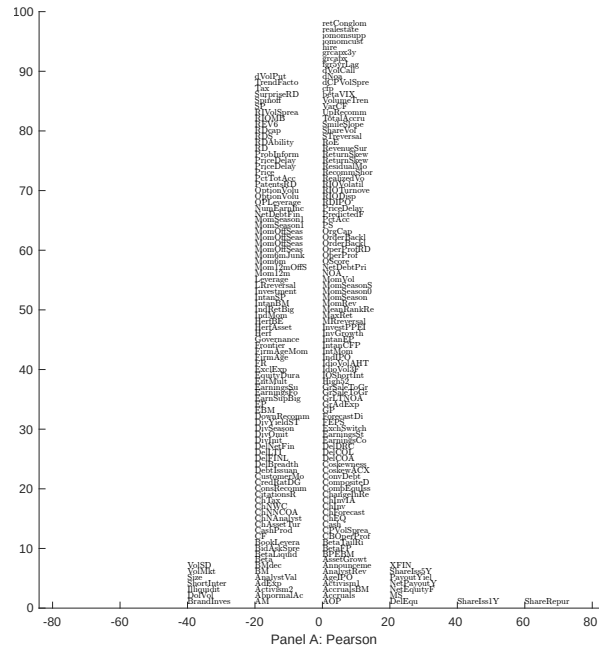


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 203 filtered anomaly signals with BDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

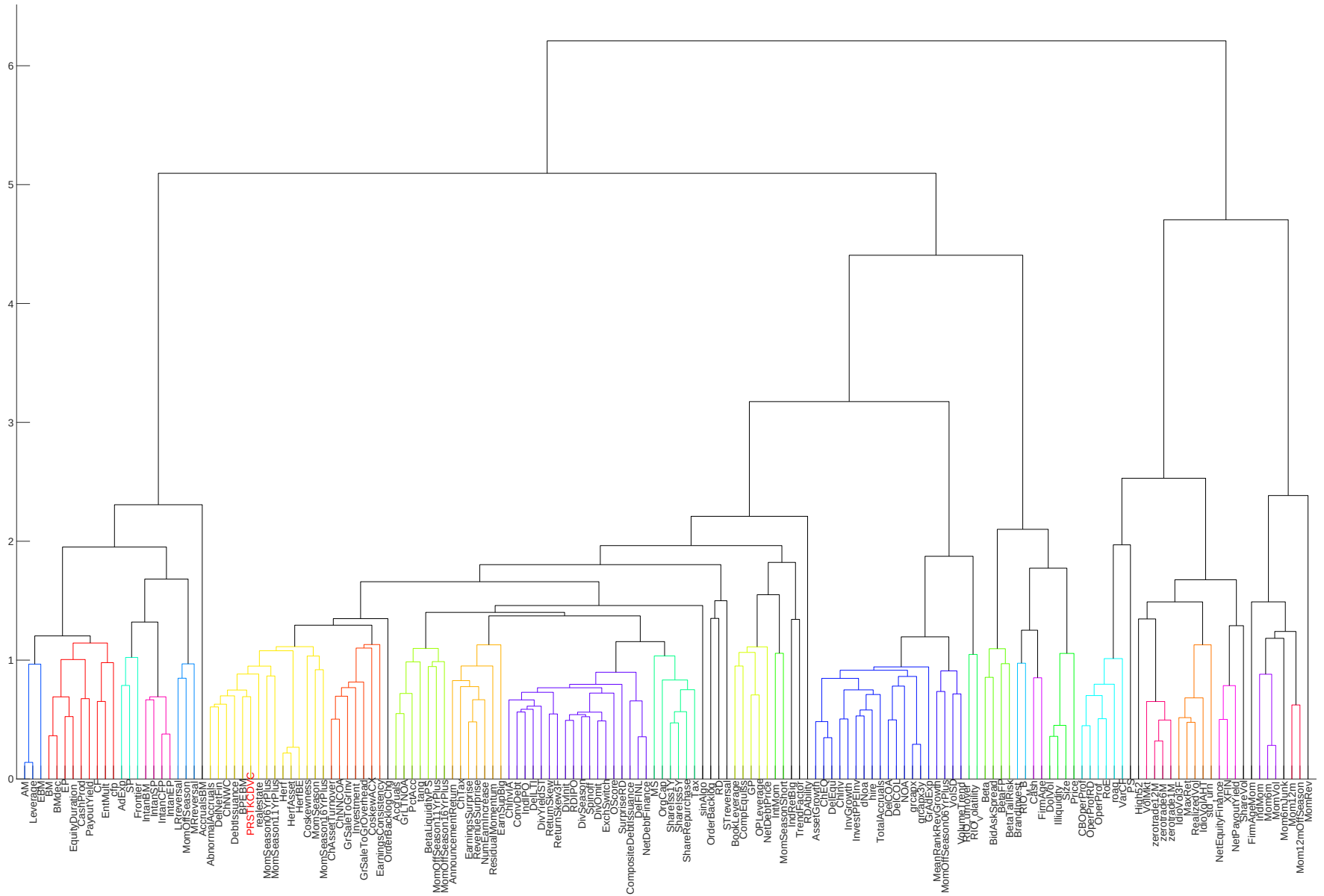


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

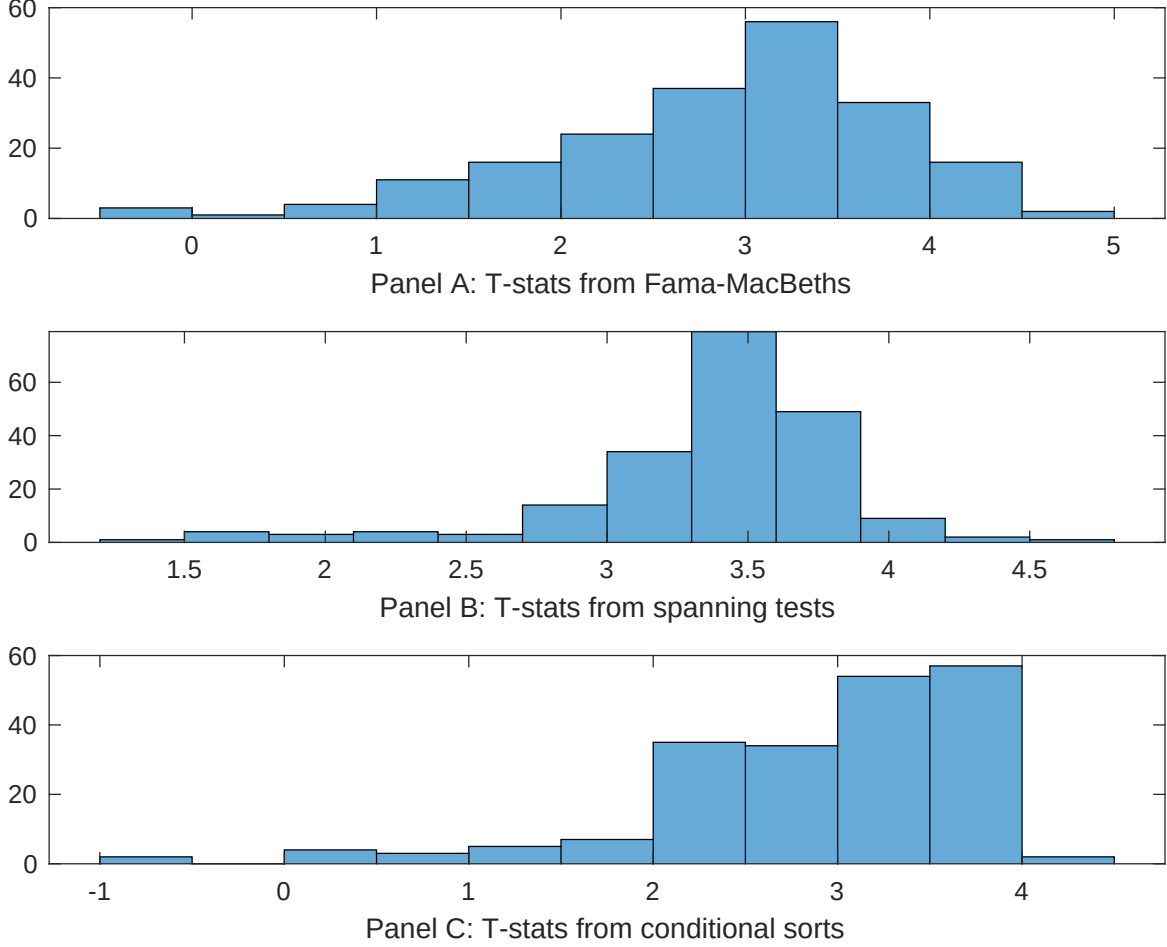


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of BDP conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDP}BDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BDP. Stocks are finally grouped into five BDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDP trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on BDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BDP}BDP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.11 [5.48]	0.12 [7.42]	0.11 [5.48]	0.12 [6.14]	0.11 [5.54]	0.11 [4.93]	0.11 [6.06]
BDP	0.27 [2.76]	0.33 [3.33]	0.26 [2.64]	0.30 [3.07]	0.26 [2.61]	0.28 [3.03]	0.29 [3.18]
Anomaly 1	0.32 [1.12]						-0.28 [-0.37]
Anomaly 2		0.28 [2.20]					0.34 [0.25]
Anomaly 3			0.39 [0.05]				0.22 [1.32]
Anomaly 4				0.30 [2.31]			0.25 [2.31]
Anomaly 5					0.16 [0.30]		0.20 [0.95]
Anomaly 6						0.17 [1.03]	0.45 [0.31]
# months	619	619	619	619	619	614	614
$\bar{R}^2(\%)$	1	2	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the BDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.38 [4.26]	0.34 [3.77]	0.40 [4.46]	0.42 [4.63]	0.38 [4.25]	0.35 [3.90]	0.28 [3.05]
Anomaly 1	-17.02 [-6.16]						-2.62 [-0.35]
Anomaly 2		-19.27 [-6.89]					-24.00 [-4.01]
Anomaly 3			-15.61 [-5.66]				9.14 [1.11]
Anomaly 4				-21.04 [-4.15]			-1.32 [-0.22]
Anomaly 5					-15.93 [-5.82]		2.32 [0.37]
Anomaly 6						-20.77 [-5.97]	-19.37 [-5.07]
mkt	0.88 [0.37]	-0.96 [-0.39]	1.33 [0.56]	3.81 [1.62]	1.11 [0.46]	3.57 [1.62]	-3.66 [-1.48]
smb	-8.05 [-2.58]	-17.77 [-5.17]	-9.83 [-3.11]	4.71 [1.09]	-12.56 [-3.87]	4.76 [1.28]	-6.18 [-1.17]
hml	-1.47 [-0.36]	0.05 [0.01]	-1.42 [-0.35]	0.56 [0.13]	-1.44 [-0.35]	-5.00 [-1.25]	0.22 [0.05]
rmw	2.83 [0.64]	3.48 [0.80]	2.90 [0.64]	-0.70 [-0.16]	2.79 [0.62]	-3.75 [-0.91]	4.22 [0.94]
cma	-5.94 [-0.96]	-6.30 [-1.03]	-7.13 [-1.15]	-13.38 [-2.17]	-7.64 [-1.23]	-6.65 [-1.07]	-1.03 [-0.17]
umd	-7.76 [-3.75]	-4.05 [-1.87]	-8.16 [-3.94]	-8.36 [-3.98]	-8.61 [-4.16]	-7.31 [-3.49]	-1.55 [-0.63]
# months	620	620	620	620	620	615	615
$\bar{R}^2(\%)$	17	18	16	14	16	16	21

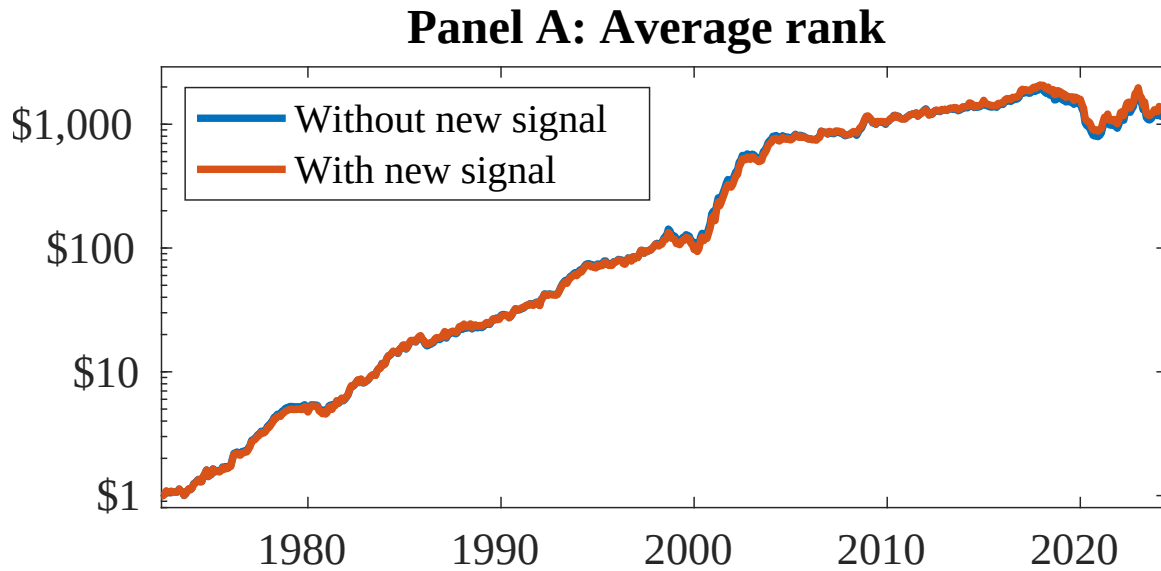


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as BDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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